

Project 1

ELECENG 2EI4 – February 8nd, 2026

Adi Ganguli - 400572384

Statement of Originality: *I hereby certify that all the work described within this report is the original work of the author. Any published (or unpublished) ideas and/or techniques from the work of others are fully acknowledged in accordance with the standard referencing practices, and I have obtained the necessary permission from the organization to submit this report*

Table of Contents

Summary	3
Design	3
Part I: Transformer	3
Part II: Rectifier	3
Part III: Filter	3
Part IV: Regulator	4
Part V: Circuit Schematic	4
Part VI: Component Calculations	4
Part VII: Expected Performance	5
Part VIII: Design Tradeoffs and Safety Considerations	5
Measurement and Analysis	7
Part I: Circuit Photograph	7
Part II: Explain the Measurement Procedure	7
Part III: Key Measurement Results	8
Part IV: Oscilloscope Output	8
Simulation	9
Part I: Circuit Schematic	9
Part II: Netlist	9
Part III: Simulation Conditions	9
Part IV: Simulation Output	10
Discussion	11
Part I: Compare the results from design, simulation, and measurement.	11
Part II: Discuss any discrepancies observed.	11
Part III: Discuss the limitations of the design and the limitations of the measurements you performed	11
Part IV: Describe any problems encountered in measurements and the troubleshooting steps you took	12

Summary

This project involves the design, simulation and implementation of a DC power supply that delivers 10 mA at $3V \pm 0.1V$ from a 120V RMS, 1 kHz Source. The main constraint of this project is that a transformer cannot be used for safety purposes, so the required transformer voltage will be determined analytically and supplied by the Analog Discovery 3 function generator. The power supply design consists of a full-wave rectifier with dual-supply and a capacitor-input filter to reduce voltage ripple, resulting in a smooth DC output.

Design

Part I: Transformer

Since a Transformer is not used, we simulate the use of a step down transformer through the input of the circuit. By determining the transformer output analytically, we can achieve a sinusoidal input that has an amplitude of 3.8V, resulting in $V_{rms} = 2.7V$. This is simulated through the AD3's Wavegen feature. To find the transformer turn ratio that would be needed, the formula for turns ratio is used,

$$\begin{aligned} \text{Turns Ratio} &= \frac{V_p}{V_s} = \frac{N_p}{N_s}, V_p = 120V \text{ rms}, V_s = 2.7V \text{ rms} \\ \text{Turns Ratio} &= \frac{120V}{2.7V} = 44.444 = 400:9 \end{aligned}$$

Therefore, the turns ratio of the theoretical transformer would be 400:9.

Part II: Rectifier

A full-wave rectifier with dual input is used to convert the AC input voltage into a unidirectional DC waveform. This topology was chosen because it utilizes both the positive and negative half-cycles to create a more accurate output voltage than a half-wave rectifier. It is also simple to physically recreate, only using two diodes. Each branch of the transformer output is attached to a single diode, resulting in a unipolar output waveform, experiencing a voltage drop of $V_d = 0.7V$.

Part III: Filter

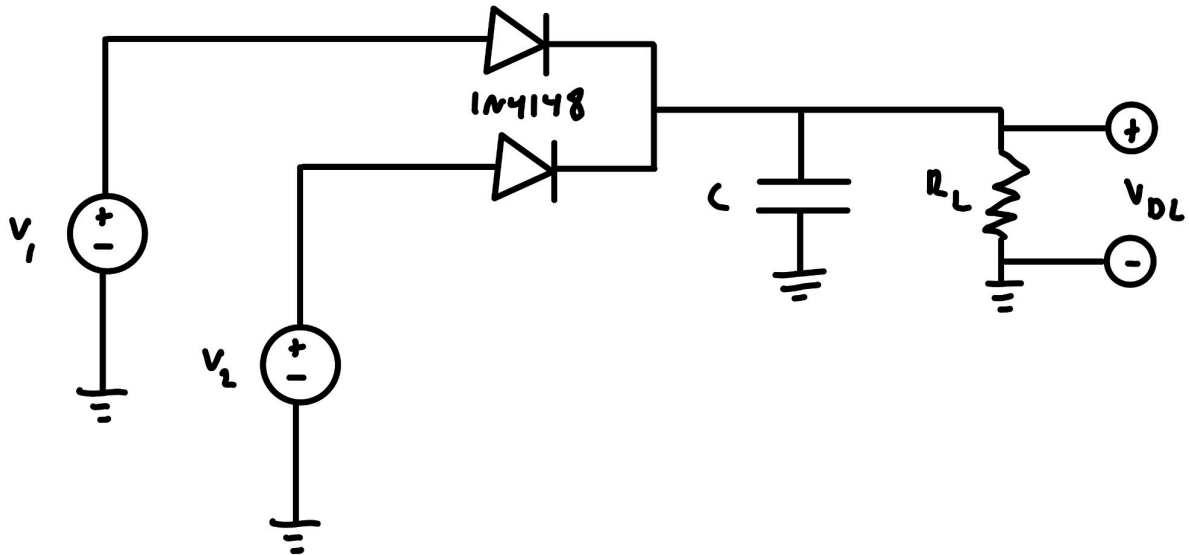
For the filter, a single capacitor-input filter was used to reduce the ripple present in the rectified output voltage. The capacitor is connected in parallel to the load, which smooths out the rectifier output and results in a small ripple that averages around the required 3V goal. For this circuit a capacitance of 100 μ F was used.

Part IV: Regulator

A regulator can be used to further smooth out the output voltage with the use of a Zener Diode. As a regulator is optional, a zener diode was not used in this circuit.

Part V: Circuit Schematic

Figure 1 - Schematic of the final circuit design



Part VI: Component Calculations

Assume that the Diodes have a voltage drop of 0.7V.

Table 1 - Calculations for the components used in the final circuit design

	Theoretical Calculation
Load Resistance R_L	To determine R_L, we identify that we need a load voltage of $3V \pm 0.1V$ and a load current of 10 mA. Using Ohm's Law: $R_{min} = \frac{V_L}{I_L} = \frac{3V-0.1V}{10mA} = 290 \Omega,$ $R_{max} = \frac{V_L}{I_L} = \frac{3V+0.1V}{10mA} = 310 \Omega$ Therefore, our resistance must be in the range of $[290,310] \Omega$. The final circuit uses a series combination of two 150 Ω resistors to make a load resistance of 300 Ω.
Capacitance C	Using equation 4.27(b) in the textbook, we note that $V_{Ripple} = \frac{I_{out}}{fC}$ where f represents frequency, and V_{Ripple} is the peak-to-peak voltage.

	Rearranging for capacitance yields $C = \frac{I_{out}}{fV_{Ripple}}$ The peak to peak voltage is the difference between the maximum and minimum allowed output voltage. $V_{pp} = V_{max} - V_{min} = 3.1V - 2.9V = 0.2V$ Since the rectifier outputs a unipolar waveform, the frequency will be doubled, so $f = 2\text{ kHz}$. $C = \frac{I_{out}}{fV_{Ripple}} = \frac{10\text{ mA}}{2\text{ kHz}(0.2\text{ V})} = 25\mu F$ Therefore a minimum capacitance of $25\mu F$ is required, but a higher capacitance can be used. The final circuit uses a $100\mu F$ capacitor.
AC Input and Transformer Turn Ratio	Assume that $V_{DC, max} = 3.1V$, and $V_D = 0.7V$. Therefore, the amplitude of the input voltage would be: $V_i = V_{DC, max} + V_D = 3.1V + 0.7V = 3.8V, V_{rms} = 2.7V$ The theoretical transformer turn ratio can now be determined: $\text{Turns Ratio} = \frac{V_p}{V_s} = \frac{N_p}{N_s}, V_p = 120V\text{ rms}, V_s = 2.7V\text{ rms}$ $\text{Turns Ratio} = \frac{120V}{2.7V} = 44.444 = 400:9$ Therefore, the turns ratio of the theoretical transformer would be 400:9.

Part VII: Expected Performance

According to calculations, the input AC signal should have an amplitude of 3.8V, and a frequency of 1 kHz. As this is a dual input, both the positive and negative waveforms will result in a full-wave rectified, pulsating DC output at twice the input frequency. With the addition of the capacitor filter, the output voltage is smoothed by reducing ripple, resulting in an approximately constant DC level around 3 V.

Part VIII: Design Tradeoffs and Safety Considerations

One major tradeoff during the measurement of the circuit was using a 300Ω resistor as a series combination of two 150Ω resistors. This is because the 2EI4 kit does not come with a native 300Ω resistor. In comparison to a 300Ω resistor, the tolerance of each 150Ω resistor

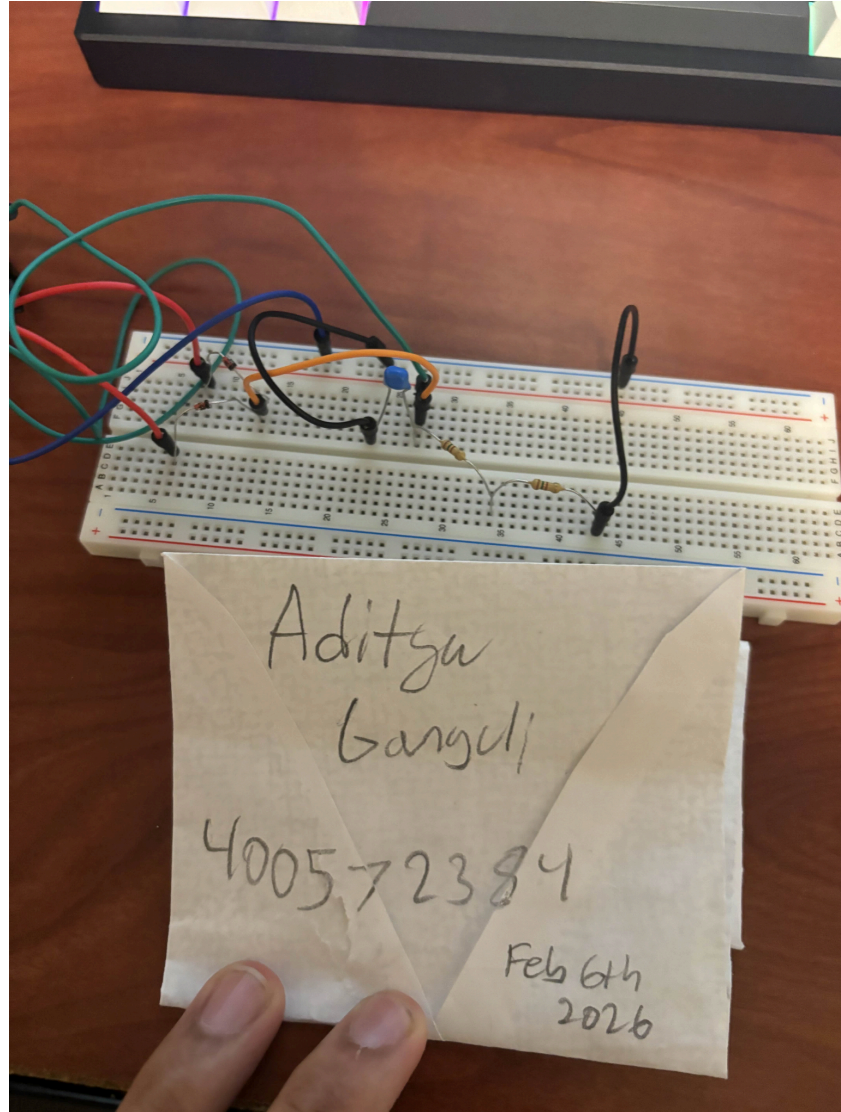
will add up, resulting in wider error and additionally increase noise contribution from the breadboard. Another tradeoff comes with the filter, as the 2EI4 kit does not come with a $25\mu F$ capacitor. However, this is the minimum capacitance required to keep the output voltage in the allowable range, meaning that a higher capacitance can be used. Therefore, the circuit uses a $100\mu F$ capacitor.

Another important consideration is the safety ratings of the components, as exceeding their power or voltage limits can pose significant hazards. The two $150\ \Omega$ resistors used to simulate the final load have a combined power rating of $0.5\ W$. Given the target output voltage of $3\ V$, the maximum power dissipated across these resistors is approximately $0.03\ W$, well within the safe operating range. Similarly, the $100\ \mu F$ capacitor (CM107) has a maximum voltage rating of $6.3\ V$, which comfortably exceeds the maximum applied voltage of $3\ V$. Finally, the 1N4148 diodes, rated for a maximum voltage of approximately $100\ V$, are easily operated safely under the circuit's maximum input voltage of $3.8\ V$.

Measurement and Analysis

Part I: Circuit Photograph

Figure 2 - Final circuit design on the breadboard



Part II: Explain the Measurement Procedure

The precise measurement of both the output voltage and current can be measured through the use of the Channel 1 pins on the AD3. Using the dual input W1 and W2 outputting two opposite sine waves, the output can now be measured using the Scope feature on the WaveForms software. To measure the voltage, the 1+ probe is connected to the node that connects the capacitor, load resistor and diodes, while the 1- probe is connected to ground. As the AD3 cannot measure current, the current is measured by creating a Custom Math Channel, and setting it to $C1/300$, which represents the measured voltage divided by the load resistance of $300\ \Omega$.

Part III: Key Measurement Results

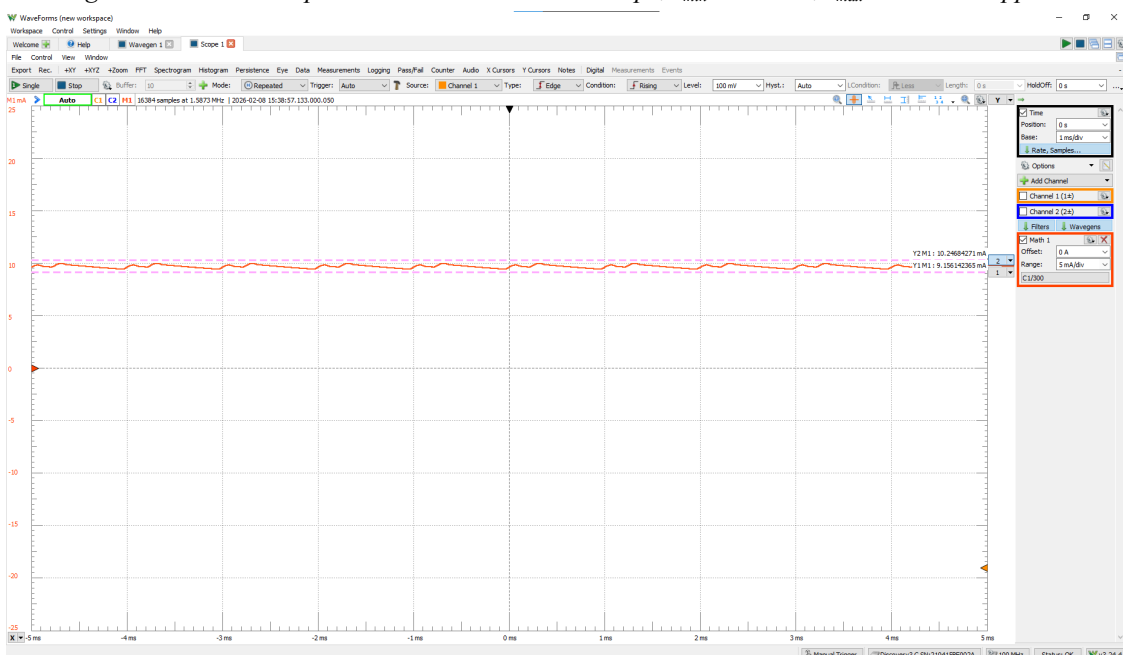
Through the use of the oscilloscope, the output voltage was mostly able to meet expectations, with a range of approximately 2.8-3.0 V, resulting in an average output voltage of 2.9 V. The waveform also shows slight ripples but mostly smoothed out with the use of the filter. As for the output current, a smoothed waveform is outputted with an average current of 9.7 mA, meeting expectations.

Part IV: Oscilloscope Output

Figure 3 - Voltage Output seen on the AD3 oscilloscope, $V_{min} = 2.8V$, $V_{max} = 3.0V$ approx.

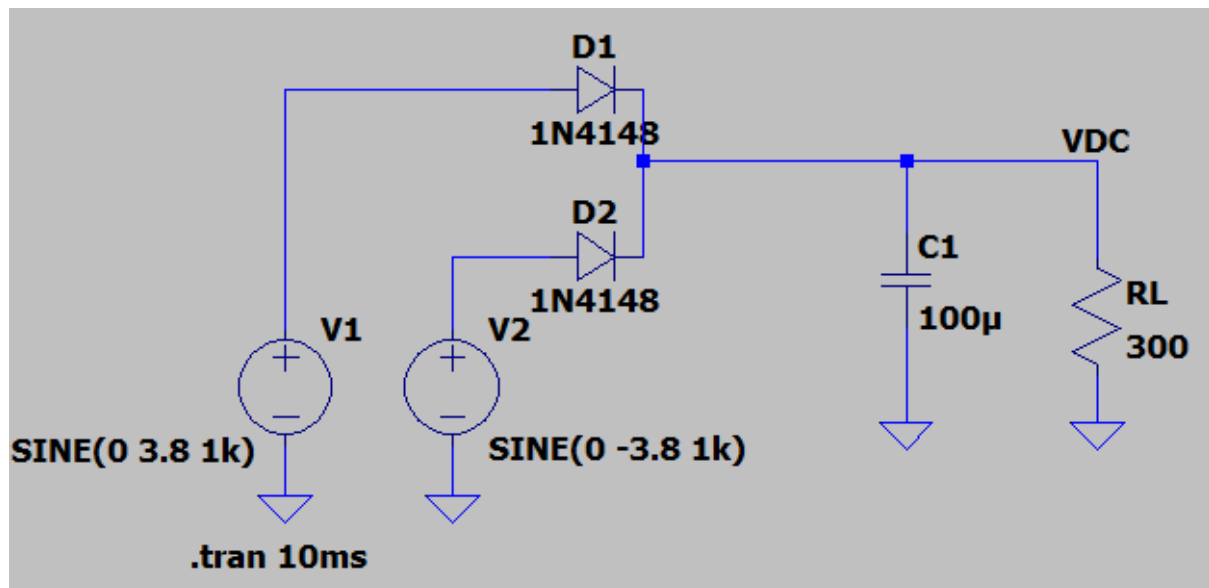


Figure 4 - Current Output seen on the AD3 oscilloscope, $I_{min} = 9.2\text{ mA}$, $I_{max} = 10.2\text{ mA}$ approx.



Simulation

Part I: Circuit Schematic



Part II: Netlist

* C:\Users\adipg\OneDrive\Documents\LTspice\2EI4 Projects\Project1.asc

* Generated by LTspice 26.0.1 for Windows.

V1 N001 0 SINE(0 3.8 1k)

D1 N001 VDC 1N4148

D2 N002 VDC 1N4148

C1 VDC 0 100μ

RL VDC 0 300

V2 N002 0 SINE(0 -3.8 1k)

.model D D

.lib C:\Users\adipg\AppData\Local\LTspice\lib\cmp\standard.dio

.tran 10ms

.backanno

.end

Part III: Simulation Conditions

The dual inputs both use a 1 kHz frequency and an amplitude of + 3.8 V and -3.8 V respectively. The diodes are the same model as the ones used in the 2EI4 kit (1N4148). The simulation is transient, with a stop time of 10ms is used for the simulation output.

Part IV: Simulation Output

Figure 5 - Voltage Output seen in the LTSpice Simulation, $V_{min} = 2.937V$, $V_{max} = 2.984V$ approx.

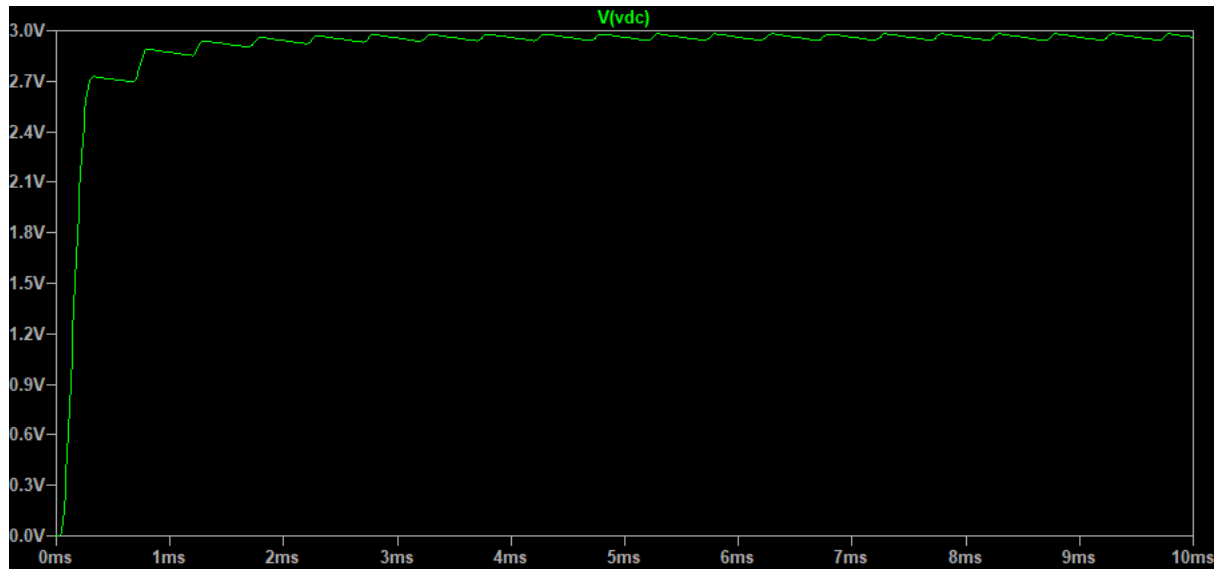
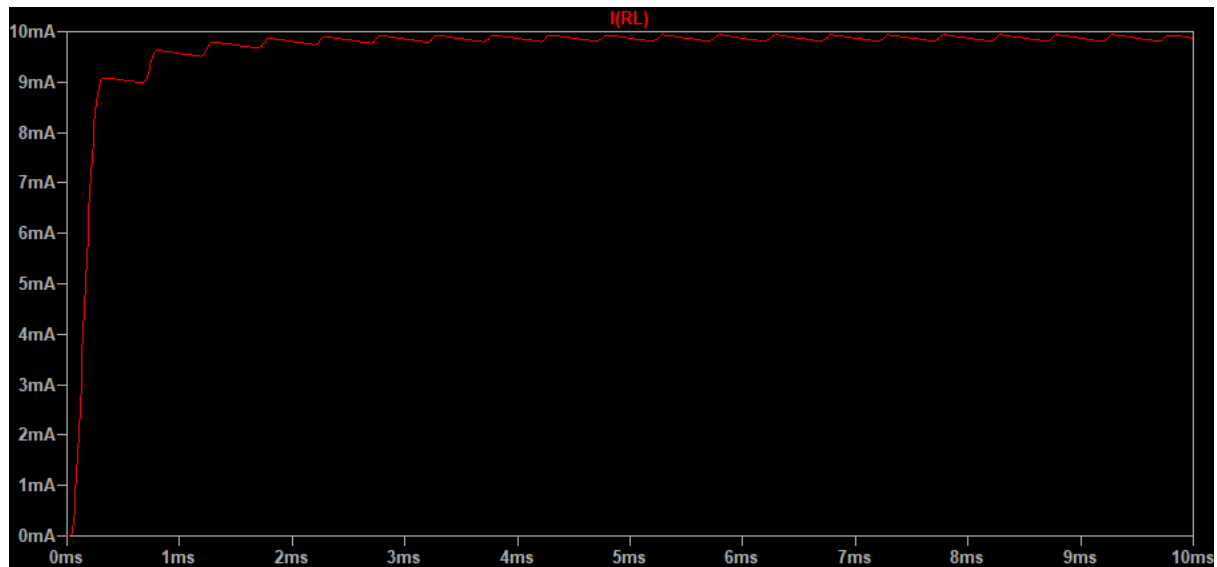


Figure 6 - Current Output seen in the LTSpice Simulation, $I_{min} = 9.80\text{ mA}$, $I_{max} = 9.97\text{ mA}$ approx.



As observed, the final output voltage is mostly smoothed out with the use of the filter, but slightly oscillates from approximately 2.94 V to 2.97 V. The load current is similar in that it is mostly smoothed out but also slightly oscillates from 9.8 mA to 10 mA.

Discussion

Part I: Compare the results from design, simulation, and measurement.

Comparing the results of the theoretical calculations, simulation and measurement show that there are some slight discrepancies between them. Notably, the measurement using the AD3 shows that an input voltage of amplitude 3.8V results with an expected output voltage. Additionally, the use of a 100 μF capacitor did not result in much change in comparison to the minimum required 25 μF . In simulation, it was found that the capacitance had little effect as long as it was at least 25 μF , with lower values such as 1 μF resulting in a more sinusoidal waveform, and higher values creating smoother output, but requiring more time to stabilize to 3V.

Part II: Discuss any discrepancies observed.

There were few discrepancies observed in the measurement of the circuit using the AD3. The final output voltage of 3V was achieved using the input amplitude of 3.8V, but was slightly lower than expected. There are multiple reasons that can contribute to this, such as internal resistance in the breadboard, wires, diodes and capacitor, which will overall affect the current throughout the circuit. Another reason can be due to the voltage drop of the diodes being slightly higher than 0.7 V, causing the voltage drop to be higher than usual. As for the current, it was mainly noted that it met the expectation of 10 mA, but slightly dipped to values as low as 9.8 mA due to the aforementioned reasons.

Part III: Discuss the limitations of the design and the limitations of the measurements you performed

Some of the limitations associated with the measurements performed can be determined as the aforementioned reasons of internal resistance in the circuit. This includes things such as the series combination of two resistors to create a load resistor, the multiple nodes used to create the circuit, resistance in the breadboard and diodes. Another limitation comes from the capacitor and its interaction with the input voltage. Although the AD3 supplies a 3.8 V, 1 kHz input, the 100 μF capacitor does not fully charge during each cycle. The limited charging time, combined with the internal diode resistance, restricts the charging current, while the capacitor discharges through the 300 Ω load between peaks. As a result, the capacitor cannot reach the ideal peak voltage, leading to ripple and a slightly reduced average DC output voltage.

Part IV: Describe any problems encountered in measurements and the troubleshooting steps you took

There was one major issue encountered with the circuit, in that the final output voltage wouldn't smooth out, causing it to look more reduced and with a lot of electrical noise changing the shape. To troubleshoot, the circuit was rewired multiple times with the same issue reappearing. The final reason that caused the noise was due to both diodes being faulty, as it was noticed that the voltage output from the diodes drastically changed to look malformed, while the input voltage remained the same. To fix this, both diodes were changed and the output looked nearly identical to the simulation, fixing the issue.